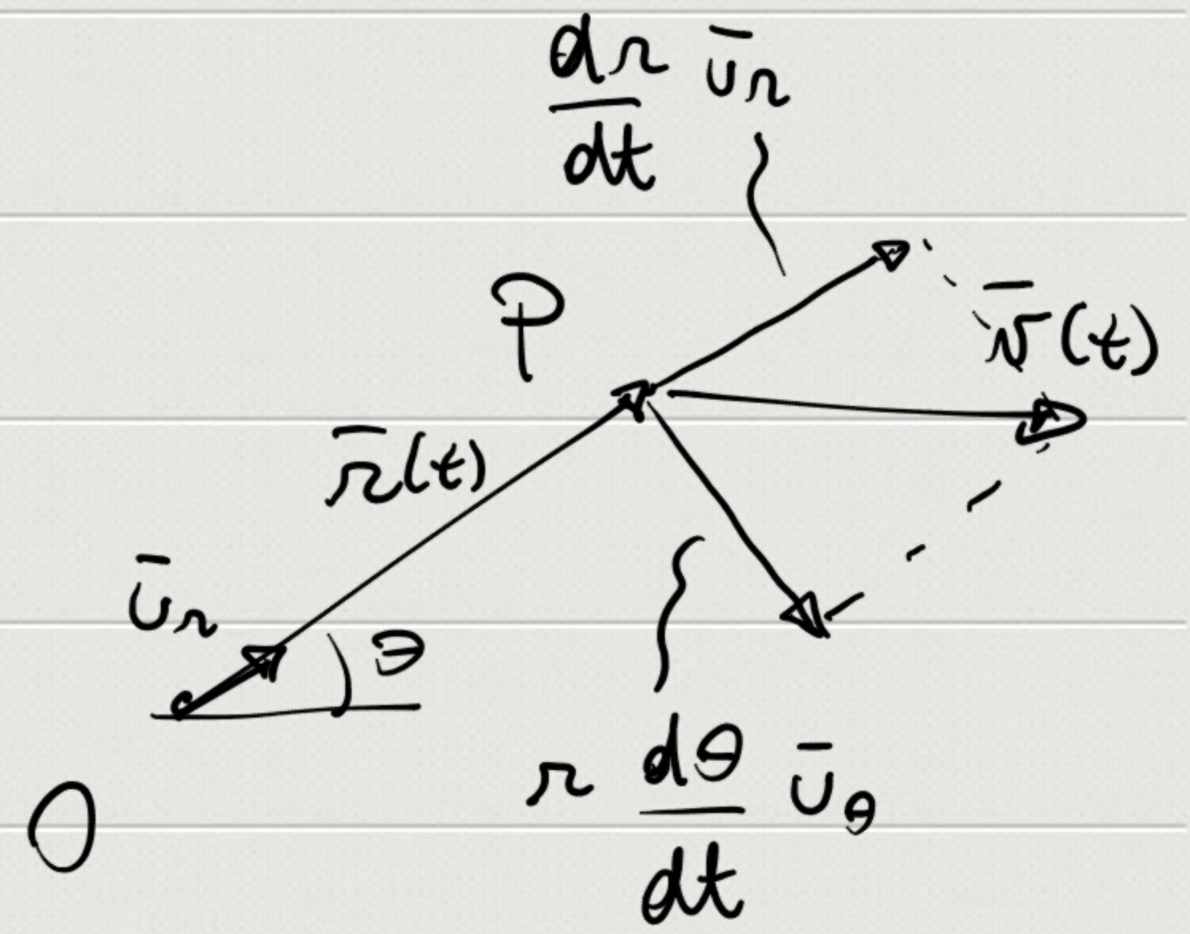


$$\bar{v}(t) = \frac{d\bar{r}}{dt} = \frac{d}{dt}\left(r(t)\,\bar{u}_r\right) = \frac{dr}{dt}\bar{u}_r + r\frac{d\bar{u}_r}{dt} =$$

$$= \frac{dr}{dt}\bar{u}_r + r\frac{d\theta}{dt}\bar{u}_\theta = \qquad \bar{u}_\theta \perp \bar{u}_r$$

$$= \bar{v}_r + \bar{v}_\theta$$

velocità radiale

velocità trasversa

$$\boxed{\bar{a}_m = \frac{\Delta \bar{v}}{\Delta t}}$$

$$\boxed{\bar{a}(t) = \frac{d\bar{v}}{dt} = \frac{d^2\bar{r}}{dt^2}}$$

$$d\bar{v} = \bar{a}(t)\,dt \Rightarrow \int_{\bar{v}_0}^{\bar{v}(t)} d\bar{v} = \int_{t_0}^{t} \bar{a}(t)\,dt$$

$$\boxed{\bar{v}(t) = \bar{v}_0 + \int_{t_0}^{t} \bar{a}(t)\,dt}$$

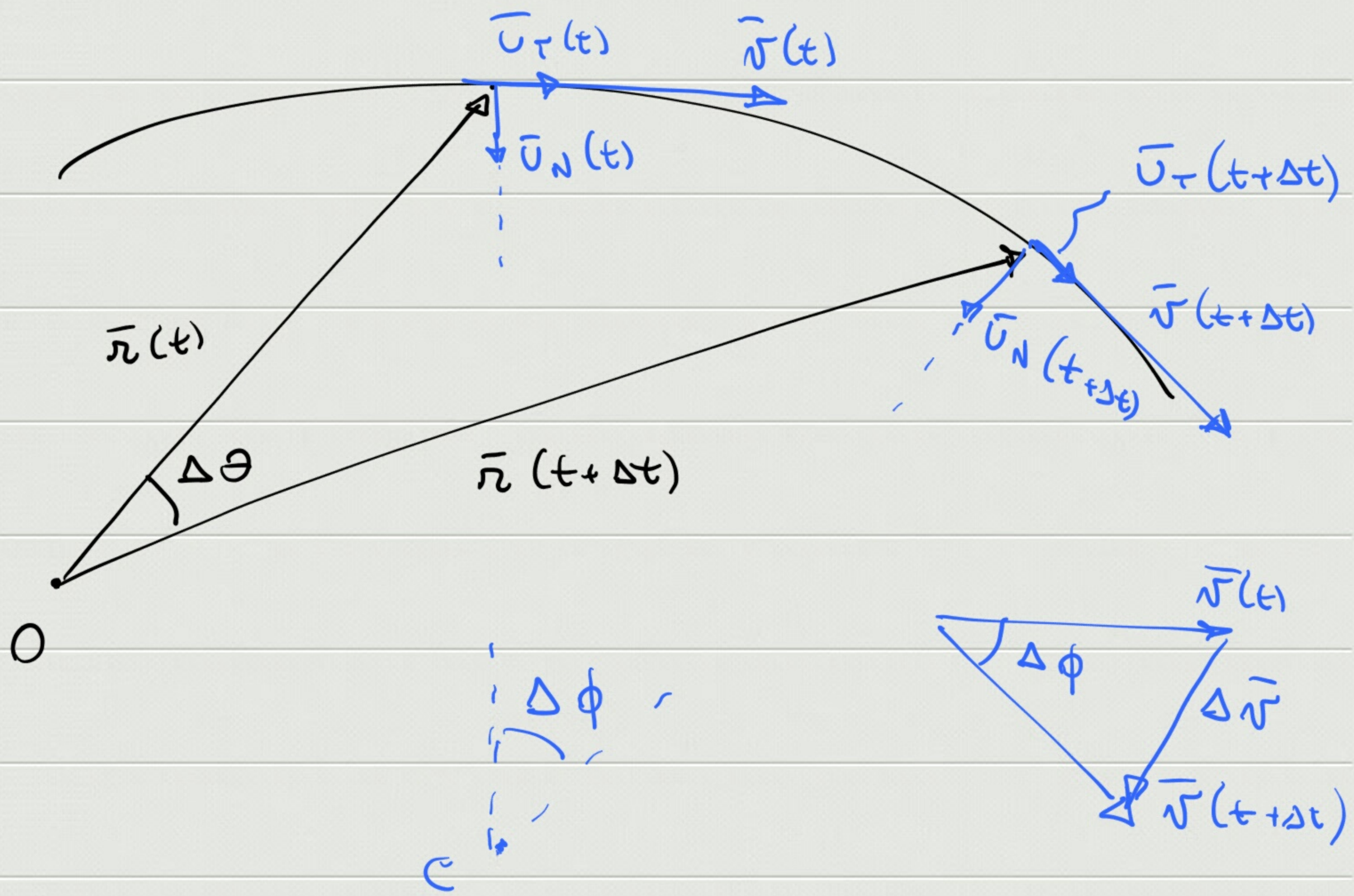
$$\bar{v}_0 = \bar{v}(t_0)$$

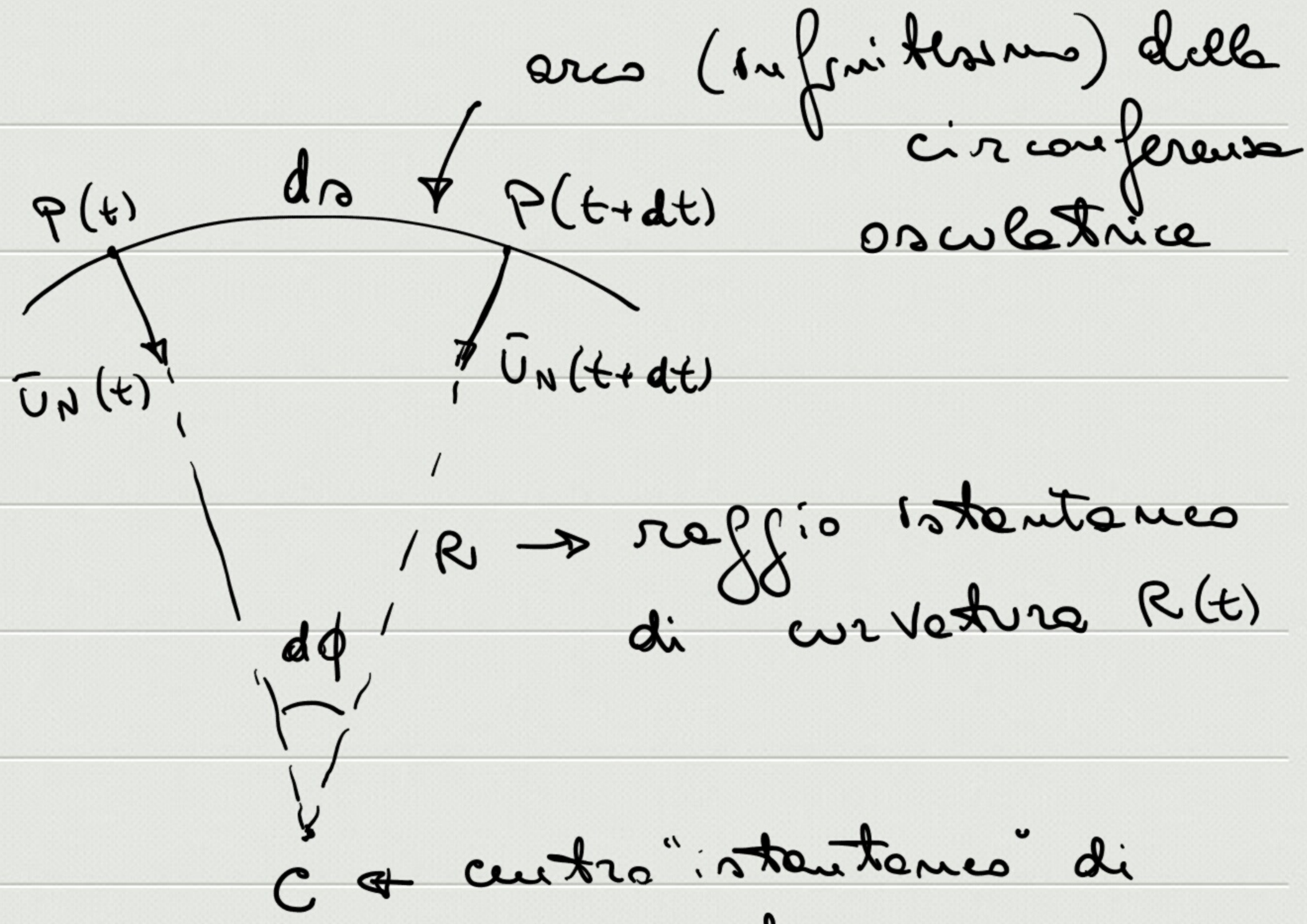
$$\bar{v} = v\,\bar{u}_T$$

$$\bar{a} = \frac{d\bar{v}}{dt} = \frac{d}{dt}\left(v\,\bar{u}_T\right) = \frac{dv}{dt}\,\bar{u}_T + v\,\frac{d\bar{u}_T}{dt} =$$

$$= \frac{dv}{dt}\,\bar{u}_T + v\,\frac{d\phi}{dt}\,\bar{u}_N = \qquad \bar{u}_N \perp \bar{u}_T$$

$$= \bar{a}_T + \bar{a}_N$$





$$ds = R\, d\phi \quad \Rightarrow \quad \frac{ds}{dt} = R \underbrace{\frac{d\phi}{dt}}_{v}$$

$$\Rightarrow \quad \frac{d\phi}{dt} = \frac{v}{R}$$

$$\bar{a}_N = v \frac{d\phi}{dt} \bar{u}_N = \frac{v^2}{R} \bar{u}_N$$

$$\Rightarrow \boxed{\bar{a}(t) = \frac{dv}{dt} \bar{u}_T + \frac{v^2}{R} \bar{u}_N = \bar{a}_T + \bar{a}_N}$$

$a_T \rightarrow$ accelerazione tangenziale

$a_N \rightarrow$ accelerazione normale o centripeta

$$\boxed{a = |\bar{a}(t)| = \sqrt{\left(\frac{dv}{dt}\right)^2 + \left(\frac{v^2}{R}\right)^2}}$$

variazione modulo $\bar{v}$

variazione direzione $\bar{v}$